

A Complete Stationary Phase Atlas for the Noisy Mean-Field Kuramoto Equation

HCS-C347 source-local reconstruction

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Abstract

For identical frequencies, positive noise, and attractive sinusoidal mean-field coupling, one theorem joins global classical probability flow, an exact free-energy dissipation law, exhaustion of stationary densities by von Mises profiles, the sharp synchronization threshold, the complete uniform Fourier spectrum, and a two-term critical expansion. The decisive Bessel quotient is proved strictly decreasing by an explicit positive-series coefficient pairing, which simultaneously closes the strict Turán inequality. Exact rational ledgers certify finite Bessel tails, stationary root brackets, and Fourier conventions; they do not replace the analytic continuum proof. No priority or arithmetic-target claim is made.

Revision certificate. Round-one Turan threshold and Fourier closure.

1 Frozen equation and main result

Let $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$, use Lebesgue measure $d\theta$, and normalize $\int_{\mathbb{T}} p d\theta = 1$. Fix $D > 0$, $K \geq 0$, and define

$$z[p] = \int_{\mathbb{T}} e^{i\theta} p(\theta) d\theta = r[p]e^{i\psi[p]}, \quad b[p](\theta) = Kr[p] \sin(\psi[p] - \theta).$$

The phase is immaterial when $r = 0$. Our equation is

$$\partial_t p = Dp_{\theta\theta} - \partial_{\theta}(b[p]p). \quad (1)$$

Theorem 1 (Noisy identical-frequency atlas). *Let $0 < \gamma < 1$ and let nonnegative $p_0 \in C^{2+\gamma}(\mathbb{T})$ have unit mass.*

- (i) *Equation (1) has a unique global classical probability solution. It is strictly positive for $t > 0$.*
- (ii) *For positive times the free energy*

$$\mathcal{F}[p] = D \int_{\mathbb{T}} p \log p d\theta - \frac{K}{2} |z[p]|^2$$

satisfies

$$\mathcal{F}[p(t)] - \mathcal{F}[p(s)] = - \int_s^t \int_{\mathbb{T}} p |\partial_{\theta}(D \log p - KC[p])|^2 d\theta d\tau, \quad (2)$$

where $C[p](\theta) = \int \cos(\theta - \phi)p(\phi)d\phi$. One may take $s = 0$ if $p_0 > 0$.

- (iii) *Every nonnegative C^2 stationary probability density is uniform, $p_* = 1/(2\pi)$, or belongs to*

$$q_{\kappa, \psi}(\theta) = \frac{e^{\kappa \cos(\theta - \psi)}}{2\pi I_0(\kappa)}, \quad \kappa = \frac{K I_1(\kappa)}{D I_0(\kappa)}. \quad (3)$$

For $K \leq 2D$, only p_ exists. For $K > 2D$, there is exactly one $\kappa > 0$, and $\psi \in \mathbb{T}$ gives its complete rotation orbit.*

- (iv) *At p_* , the constant mass mode has eigenvalue zero, the real first-harmonic plane has eigenvalue $K/2 - D$, and the real harmonic plane of order $n \geq 2$ has eigenvalue $-Dn^2$. The constant direction is removed on the probability tangent space.*

- (v) *If $\delta = K/D - 2 \downarrow 0$ along the nonuniform branch, then*

$$\kappa^2 = 4\delta + \frac{2}{3}\delta^2 + O(\delta^3), \quad r^2 = \delta - \frac{5}{6}\delta^2 + O(\delta^3).$$

2 Global probability flow and dissipation

The map $p \mapsto b[p]$ uses only two first Fourier coefficients and is smooth and locally Lipschitz between the relevant Hölder spaces. Standard heat-semigroup iteration gives a unique local classical solution. Mass conservation follows by integrating (1); comparison preserves nonnegativity, and the strong maximum principle gives strict positivity for positive time.

This local solution cannot blow up at finite time. For probability densities, $\|\partial_\theta^j b[p]\|_\infty \leq K$ for every j . At a spatial maximum of

$$p_t = Dp_{\theta\theta} - b[p]p_\theta - (\partial_\theta b[p])p$$

we obtain $\partial_t \max p \leq K \max p$, whence $\|p(t)\|_\infty \leq e^{Kt} \|p_0\|_\infty$ on finite intervals. Periodic parabolic Schauder estimates now give continuation through every finite time.

Because $\partial_\theta C[p] = b[p]/K$, with the zero-coupling case read directly, equation (1) is

$$p_t = \partial_\theta [p \partial_\theta (D \log p - KC[p])]. \quad (2)$$

The first variation of $-K|z[p]|^2/2$ is $-KC[p]$. Multiplying (2) by the chemical potential and integrating by parts proves (2). Starting at $s > 0$ handles initial zeros.

3 Stationary-flux exhaustion

A nonnegative stationary C^2 density is positive by the elliptic strong maximum principle. Its flux

$$J = b[p]p - Dp_\theta$$

is constant. Division by p and one circuit around $\mathbb{T}$ give

$$0 = \int \frac{p_\theta}{p} = \frac{1}{D} \int b[p] - \frac{J}{D} \int \frac{d\theta}{p(\theta)}.$$

The sine drift integrates to zero and the last integral is positive, so $J = 0$. Therefore

$$\partial_\theta \log p = \frac{Kr}{D} \sin(\psi - \theta) = \partial_\theta \left(\frac{Kr}{D} \cos(\theta - \psi) \right).$$

Normalization gives the von Mises form in (3). Conversely its first moment is

$$z[q_{\kappa, \psi}] = e^{i\psi} R(\kappa), \quad R(\kappa) = I_1(\kappa)/I_0(\kappa),$$

so zero flux is equivalent to $\kappa = (K/D)R(\kappa)$. This proves the stationary list in both directions.

4 Strict Turán inequality and the phase threshold

Lemma 1. *The function $q(\kappa) = I_1(\kappa)/(\kappa I_0(\kappa))$ decreases strictly from $1/2$ to zero on $(0, \infty)$. Equivalently,*

$$I_1(\kappa)^2 - I_0(\kappa)I_2(\kappa) > 0. \quad (3)$$

Proof. Put $x = \kappa^2$, $A(x) = I_1(\sqrt{x})/\sqrt{x}$, and $B(x) = I_0(\sqrt{x})$. Their positive entire coefficients satisfy

$$A = \sum_{m \geq 0} a_m x^m, \quad B = \sum_{m \geq 0} b_m x^m, \quad \frac{a_m}{b_m} = \frac{1}{2(m+1)}.$$

Pairing indices in the derivative numerator yields

$$A'B - AB' = \sum_{m > n} (m-n)(a_m b_n - a_n b_m) x^{m+n-1} < 0. \quad (4)$$

Thus $q = A/B$ is strictly decreasing and $q(0+) = 1/2$. Moreover $0 < R(\kappa) < 1$, since it is the strict expectation of $\cos \theta$ under a positive von Mises density, so $q(\kappa) < 1/\kappa \rightarrow 0$.

Termwise Bessel recurrences give $R' = 1 - R^2 - R/\kappa$ and $I_2 = I_0 - 2I_1/\kappa$. Consequently $q' < 0$ is equivalent to $\kappa(1 - R^2) < 2R$, and multiplication by I_0^2/κ gives (3). Hence the strict Turán inequality is proved rather than assumed. $\square$

For $a = K/D > 0$, a nonzero stationary solution obeys $q(\kappa) = 1/a$. Lemma 1 gives no solution for $a \leq 2$ and exactly one for $a > 2$, completing the transition proof.

5 Complete uniform Fourier linearization

Write $p = p_* + \varepsilon u$, $\int u = 0$. The linearization is

$$Lu = Du_{\theta\theta} - p_* \partial_\theta b[u], \quad b[u] = K \operatorname{Im}(z[u]e^{-i\theta}). \quad (5)$$

For $u = a \cos \theta + c \sin \theta$, $z[u] = \pi(a + ic)$, and the last term in (5) is $Ku/2$. Every harmonic of order $n \geq 2$ has zero first moment, leaving $-Dn^2$. Constants give mass conservation. This proves every eigenvalue and real multiplicity stated in Theorem 1.